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 SURVEY

Voice Cloning: A Survey of Zero-Shot and Controllable Speech Synthesis

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ABSTRACT This survey examines zero-shot voice cloning through the linked views of representation, generation, control, and deployment. Rather than treating recent systems as isolated milestones, we organize the literature around the technical decisions that shape modern speaker-conditioned synthesis: neural codec design, acoustic-token modeling, autoregressive and non-autoregressive decoding, diffusion and flow-matching objectives, multilingual conditioning, streaming constraints, preference alignment, controllability, and safety. We review representative systems including YourTTS, VALL-E, Voicebox, F5-TTS, MaskGCT, Seed-TTS, CosyVoice 2, MiniMax-Speech, GLM-TTS, Spark-TTS, and Qwen3-TTS, and we compare how these systems trade off naturalness, speaker similarity, latency, controllability, and deployment risk. The survey is unified by a single frame, which is a deployment-oriented reading of the codec language model era. That frame is delivered through four contributions. We provide a four-axis taxonomy across representation, generation, control, and deployment. We provide a qualitative architectural comparison of representative systems along axes that headline metrics obscure. We provide a structured treatment of evaluation reliability that maps automatic and human metrics to the quality dimensions they actually capture. We provide a consolidated safety and governance perspective that couples technical mechanisms with dataset licensing and a pre-deployment checklist. Three field-level shifts recur across these views. The first is the move from waveform or spectrogram prediction toward discrete acoustic-token generation. The second is the transition from offline quality-first models toward streaming and conversational architectures. The third is the growing need for trustworthy evaluation, watermarking, consent-aware deployment, and demographic robustness. We conclude by identifying open problems in benchmark standardization, speaker-similarity assessment, multilingual low-resource performance, preference-aligned synthesis, and secure real-world use.

INDEX TERMS Zero-shot voice cloning, controllable speech synthesis, neural audio codecs, codec language models, multilingual text-to-speech, streaming speech synthesis, speech generation, survey.

I. INTRODUCTION

Voice cloning from a short reference utterance has become one of the most visible capabilities of modern neural speech synthesis. A system that previously needed speaker-specific recordings and adaptation can now infer speaker characteristics from a brief prompt and synthesize new linguistic content in the same perceived voice. This change has been enabled by progress in neural audio codecs, sequence modeling, self-supervised speech representations, diffusion and flow-based generation, and large-scale multilingual training [1], [2], [3], [4], [5], [6].

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The focus of this survey is the zero-shot and controllable setting. Unlike conventional speaker adaptation, zero-shot voice cloning (ZS-VC) requires the model to extract speaker identity from a very short sample, preserve that identity under unseen text, maintain intelligibility and natural prosody, and increasingly meet latency requirements suitable for interactive use. These requirements make ZS-VC a system problem as much as a modeling problem: representation, conditioning, generation strategy, evaluation protocol, and deployment constraints all interact.

Earlier surveys provide useful foundations for neural TTS and voice cloning [1], [2]. Our contribution is narrower and more recent: we analyze the recent wave in which codec language models, flow matching, masked-codec prediction,

streaming-first architectures, preference alignment, multi-modal conditioning, and safety mechanisms became central. The aim is not only to list systems, but to explain why specific design choices emerged and how they changed the field's assumptions.

A useful way to understand the period is through successive design pressures. YourTTS and EnCodec showed that multilingual speaker-conditioned synthesis and neural codec representations could serve as practical foundations [3], [7]. VALL-E and Voicebox then reframed speech generation around codec tokens and flexible generative objectives [8], [9]. Later systems such as F5-TTS, MaskGCT, Seed-TTS, CosyVoice 2, Fish-Speech, and StyleTTS 2 pushed the field toward stronger quality, faster inference, and richer control [10], [11], [12], [13], [14], [15]. The most recent line of work extends the problem toward preference/RL alignment, explicit emotion and duration control, full-duplex interaction, multimodal synthesis, and secure deployment [16], [17], [18], [19], [20].

Two prior surveys sit closest to this one. Tan et al. [1] covers the broader neural TTS landscape through 2021 but predates the codec language model era, streaming architectures, and preference-based post-training. Azzuni and El Saddik [2] emphasizes voice-cloning security through 2024 but does not develop a technical taxonomy of the codec language model systems that now dominate the field. This survey occupies the deployment-oriented gap between them.

All four contributions of this survey are unified by a single question. What makes a zero-shot voice cloning system usable, safe, and comparable in a real deployment, not only what makes it score well on a benchmark. The specific contributions are as follows.

- 1) **A deployment-oriented technical taxonomy of the codec-LM era.** We organize modern zero-shot voice cloning around four linked axes, namely representation, generation, control, and deployment, and use them to explain why particular design choices emerged rather than only that they were adopted.
- 2) **A qualitative architectural comparison** of representative systems along dimensions that headline metrics obscure: architecture family (AR, NAR, flow, or hybrid), input representation, speaker-conditioning mechanism, codec type, controllability features, and open-source status (Table 2). This complements the quantitative view in Table 4.
- 3) **A structured treatment of evaluation reliability**, mapping automatic and human metrics to the quality dimensions they actually capture (naturalness, similarity, intelligibility, prosody) and identifying where automatic scores diverge from listener judgment (Table 3).
- 4) **A consolidated safety and governance perspective** that couples technical mechanisms (watermarking, anti-spoofing, unlearning, bias auditing) with dataset licensing, speaker rights, and a concrete pre-deployment checklist for practitioners (Table 5).

TABLE 1. Positioning of this survey relative to the most closely related prior reviews. "ZS-VC" is zero-shot voice cloning. "✓" means covered in depth, "∼" means partial or peripheral coverage, and "✗" means not covered.

Dimension	Tan et al. [1]	Azzuni & El Saddik [2]	Our survey
Primary focus	Neural TTS	VC security	ZS-VC systems
Time window	–2021	–2024	2022–2026
Codec-LM systems	✗	∼	✓
Streaming and real-time	∼	✗	✓
Alignment and RL post-training	✗	✗	✓
Controllability taxonomy	∼	∼	✓
Multilingual and low-resource	∼	∼	✓
Evaluation reliability	∼	∼	✓
Safety and speaker rights	∼	✓	✓
Deployment checklist	✗	✗	✓

Table 1 positions this survey relative to the two most closely related prior reviews. Throughout the body, each system subsection includes an explicit trade-off statement so that the survey delivers comparative judgment rather than a linear enumeration of features. Supplementary resources, including an interactive timeline and a searchable bibliography of surveyed systems, are provided as companion aids to this survey.¹

A. ZS VOICE CLONING EVOLUTION ANALYSIS

The recent trajectory of zero-shot voice cloning is best viewed as overlapping methodological phases rather than a single linear race. The foundation phase established the ingredients that later systems reused at scale. YourTTS demonstrated that multilingual zero-shot multi-speaker synthesis could be implemented in a unified VITS-style framework, while EnCodec provided a high-fidelity neural codec whose discrete representations became natural targets for token-based generation [3], [7]. LoRA introduced parameter-efficient adaptation mechanisms that are relevant to personalization and lightweight deployment, and work on over-smoothing clarified why deterministic speech generators often lose expressive detail [21], [22].

The next phase can be described as the codec-language-model turn. VALL-E made the case for predicting codec tokens autoregressively from text and a short speaker prompt, thereby connecting speech synthesis with the broader logic of language modeling [8]. Voicebox expanded the design space by using flow matching for non-causal refinement and editing-style generation [9]. HierSpeech++ pursued semantic–acoustic separation through self-supervised representations, while Mega-TTS and Mega-TTS 2 introduced more structured prompt conditioning [23], [24], [25]. These systems did not merely improve metrics; they changed what

¹Timeline: <https://ntiwari78.github.io/zsvoicecloning/timeline.html> Bibliography: <https://ntiwari78.github.io/zsvoicecloning/refs.html> An archival snapshot of these resources will be deposited on Zenodo at the time of publication.

researchers considered the basic object of synthesis: not a waveform alone, but a controllable sequence of acoustic representations.

A subsequent wave emphasized architectural efficiency and scalable generation. F5-TTS simplified non-autoregressive flow matching by removing explicit duration and phoneme-alignment modules [10]. MaskGCT used masked codec prediction to exploit bidirectional context across codec levels [11]. Seed-TTS combined autoregressive coarse generation, diffusion refinement, self-distillation, and post-training alignment [12]. CosyVoice 2, Fish-Speech, and StyleTTS 2 showed that practical systems could pursue low latency, multilingual coverage, and expressive control simultaneously, although they made different trade-offs among these goals [13], [14], [15].

In the recent times, the field's center of gravity shifted from pure synthesis quality toward controllability, alignment, evaluation rigor, and deployment trust. GLM-TTS applies Group Relative Policy Optimization to speech generation, IndexTTS2 separates emotion and duration control more explicitly, Spark-TTS reduces acoustic sequence length through decoupled tokenization, and ClonEval, DisCo-Speech, and VoiceMark focus on evaluation, representation separation, and misuse mitigation [16], [17], [20], [26], [27], [28].

The newest frontier is streaming-first and multimodal. Qwen3-TTS uses a dual-track architecture for very low first-packet latency, PersonaPlex targets controlled conversational synthesis, MM-Sonate integrates multimodal and dialogue context, Fish-Speech S2 improves multilingual streaming, and MiniMax-Speech combines a learnable speaker encoder with a Flow-VAE design [18], [19], [29], [30], [31]. In aggregate, these phases show a field moving from feasibility to deployability: the central question is no longer only whether a voice can be cloned, but whether it can be cloned controllably, safely, quickly, fairly, and under realistic interaction constraints.

II. SCOPE AND SELECTION METHODOLOGY

To keep the survey reproducible and to make its coverage boundaries explicit, we followed a systematic, PRISMA-inspired search and screening process. We queried four literature sources, namely arXiv, IEEE Xplore, the ACL Anthology, and Semantic Scholar, over the window January 2022 to February 2026, complemented by backward and forward citation chasing from key system papers. Semantic Scholar was used to catch cross-venue and newly indexed records that had not yet propagated to the three primary sources at the time of the search. The search combined terms for the task ("zero-shot", "voice cloning", "speaker adaptation", "voice conversion"), the representation and model class ("neural codec", "codec language model", "acoustic token", "flow matching", "diffusion", "masked generative"), and the deployment axis ("streaming", "real-time", "controllable", "expressive",

"watermarking"), joined disjunctively within each group and conjunctively across groups.

Records were screened in three stages. An initial deduplicated pool of approximately 312 records was reduced to 94 after title/abstract screening removed off-topic hits (e.g., speech enhancement, ASR-only work, singing-voice synthesis) and clearly out-of-scope voice conversion. Screening was performed by the first author with disagreements at the eligibility stage resolved through discussion with the co-authors, and no formal inter-rater statistic was computed. The system-level works were selected using three criteria: (1) the work addresses zero-shot or few-shot voice cloning with ≤ 30 seconds of reference audio; (2) it introduces a non-trivial architectural, training, evaluation, or deployment contribution; and (3) the results are reproducible through open-source code, detailed technical reporting, or peer-reviewed publication. We exclude voice conversion systems that require parallel data or speaker enrollment, singing voice synthesis, and pure TTS systems without speaker conditioning.

The PRISMA-style flow of the literature search and screening process proceeded through four stages, summarized below (counts are approximate and reflect the state of the corpus at submission):

- 1) **Identification.** 312 deduplicated records were identified from arXiv, IEEE Xplore, the ACL Anthology, and Semantic Scholar, spanning January 2022 to February 2026. Approximately 12 official technical reports and code releases (Seed-TTS, MiniMax-Speech, CosyVoice, and related tech reports) were added at this stage and are included in the 312 count. Duplicates were removed inside the identification stage so this count already reflects deduplicated records.
- 2) **Title and abstract screening.** 94 records were retained. Approximately 218 records were removed at this stage as off-topic (ASR-only work, speech enhancement, singing-voice synthesis) or as clearly out-of-scope voice conversion.
- 3) **Full-text eligibility.** The 94 retained records were assessed against the three inclusion criteria.
- 4) **Included.** This final set comprises the zero-shot voice cloning systems reviewed in the body together with the methodological references (Transformer, diffusion, flow matching, codec, and evaluation-tool papers) that recur across them.

III. TECHNICAL BACKGROUND AND FOUNDATIONS

A. FROM CLASSICAL VOICE CONVERSION TO ZERO-SHOT CLONING

Although parallel-data voice conversion is excluded from the systems surveyed in later sections, we include it briefly here as historical context because modern zero-shot cloning inherits its representational vocabulary and its speaker-identity framing from a longer line of voice conversion (VC) research. Its core problem, transferring a target speaker's identity onto new content, comes directly from that line, and a brief bridge

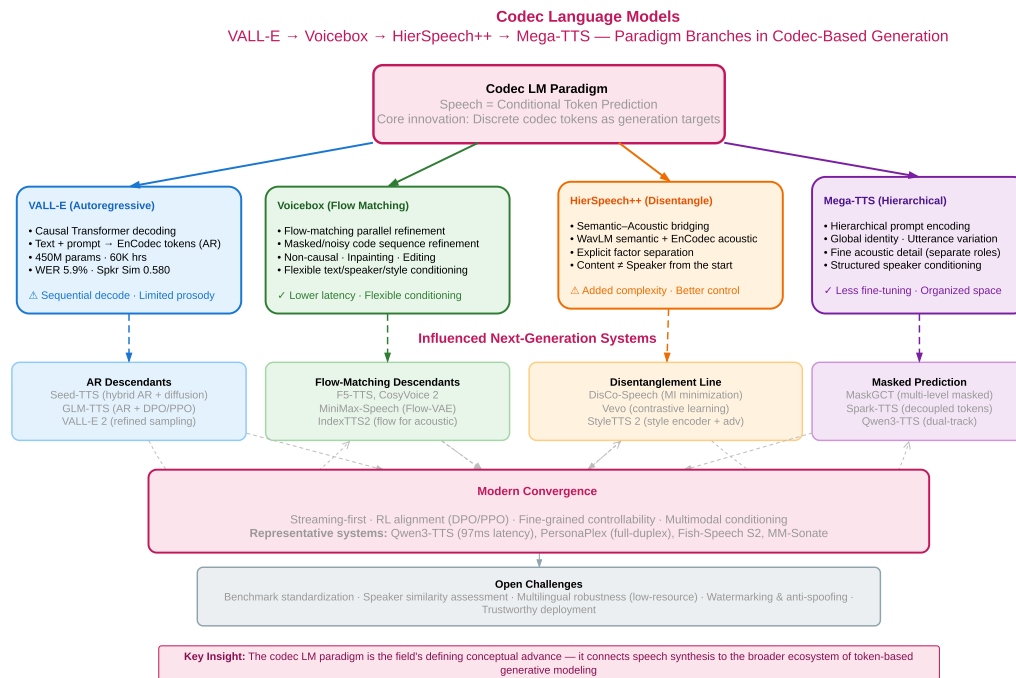


FIGURE 1. Conceptual evolution of codec-based zero-shot voice cloning. The diagram groups representative systems into autoregressive, flow-based, disentanglement-oriented, and hierarchical prompt-conditioning families, highlighting how later work recombines these ideas for streaming, alignment, and multimodal deployment [8], [9], [23], [24].

clarifies which ideas were genuinely new after 2022 and which were reformulations of older ones. Early statistical VC posed conversion as a mapping between parallel source and target features. Gaussian-mixture-model (GMM) approaches learned a joint density over aligned spectral frames and converted via probabilistic regression [32], later refined by maximum-likelihood trajectory estimation with global-variance compensation to counter the over-smoothing that made converted speech sound muffled [33]. These methods established the representational vocabulary, namely spectral envelopes, alignment, and prosody transfer, but required parallel training data and produced audibly artificial results.

The next stage replaced the GMM mapping with deep neural networks. Feed-forward and recurrent networks learned nonlinear spectral mappings that removed some of the muffling caused by GMM regression [34], and DNN-based conversion coupled with high-quality continuous vocoders further narrowed the naturalness gap relative to the GMM era [35]. In parallel, the community moved from per-pair conversion toward speaker adaptation and speaker embedding: fixed-dimensional identity vectors such as x-vectors [36] and end-to-end verification embeddings [37] allowed a single synthesis model to be conditioned on a compact speaker representation rather than retrained per target. Transfer learning from speaker-verification encoders into multi-speaker TTS made this conditioning practical, enabling synthesis of voices unseen during TTS training from only a few seconds of reference audio [38]. This embedding-conditioning step is the conceptual hinge between classical VC and modern

practice: once identity could be supplied as a prompt-derived vector, the remaining advances, namely neural codec targets, token-based generation, and flow matching, extended how the conditioned content is produced rather than whether zero-shot transfer was possible at all. This is a compressed history intended as a bridge, not a comprehensive VC survey; the works cited above are chosen for their direct influence on the speaker-conditioning interfaces used in Section IV onward. The systems surveyed below build directly on this foundation.

Having established the terminology and historical framing, we now review the technical foundations that recur throughout modern zero-shot voice cloning systems. These foundations are important because many contemporary models differ less in their high-level objective than in the specific combination of codec design, tokenization strategy, conditioning mechanism, and generative architecture they employ.

B. NEURAL AUDIO CODECS

Modern zero-shot voice cloning depends heavily on learned audio representations that are both compact and information preserving. Neural codecs play this role by mapping waveform audio into structured latent spaces that can either remain continuous or be discretized for token modeling. Compared with conventional signal-processing codecs, neural codecs are optimized directly for perceptual reconstruction and downstream modeling utility. *EnCodec* [3] introduced a modern neural audio compression

framework:

$$x \rightarrow \text{Encoder}(x) \rightarrow \{\mathbf{q}_t\}_t$$

(quantized codes at ~ 50 Hz frame rate) (1)

Its importance lies not only in its reconstruction quality but also in the fact that its discretized outputs can be treated as modeling targets. EnCodec uses learned residual vector quantization (RVQ), adversarial training, and scalable bandwidth control, thereby making it suitable both for codec reconstruction and for sequence modeling. In later codec-language-model systems, these quantized codes became the basic units of generation. Although EnCodec relies on learned vector quantization, more recent work has shown that simpler discretization strategies can be advantageous. *Finite Scalar Quantization (FSQ)* [39] replaces learned codebooks with fixed scalar levels:

$$q_i(x_i) = \text{Round}\left(\frac{x_i}{s}\right) \cdot s,$$

$$s = \frac{1}{L}. \quad (2)$$

This formulation avoids certain collapse modes associated with vector-quantized models, simplifies optimization, and often improves training stability. Its influence is visible in later systems such as MaskGCT and Fish-Speech, which adopt scalar-quantization-inspired representations to improve robustness.

C. DISCRETIZATION AND THE CODEC LM PARADIGM

The decisive conceptual step in recent zero-shot TTS was to treat codec outputs as a speech analogue of language-model tokens. In systems such as VALL-E, this allows speaker prompting and text conditioning to be expressed as conditional acoustic-token prediction [8]:

$$P(\mathbf{c} \mid \text{text}, \mathbf{c}_{\text{ref}}) = \prod_{t=1}^T P(c_t \mid c_{<t}, \text{text}, \text{Prompt}(\mathbf{c}_{\text{ref}})). \quad (3)$$

Here, $\mathbf{c}$ denotes the target acoustic code sequence and $\mathbf{c}_{\text{ref}}$ denotes the prompt codes extracted from the reference speaker. Throughout the paper we follow the convention that $\mathbf{c}_*$ refers to discrete codec or token sequences, while $\mathbf{x}_*$ refers to raw or continuous waveform/feature representations. This formulation is important because it makes modern TTS compatible with advances in large-scale token modeling, including causal language modeling, masked prediction, and flow-based sequence generation [8], [9], [11].

D. ATTENTION AND TRANSFORMER FOUNDATIONS

The Transformer [4] remains the dominant backbone for modern zero-shot voice cloning:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V. \quad (4)$$

In this context, the Transformer provides several essential capabilities. It allows text representations to be contextualized over long sequences, supports rich cross-attention between text and acoustic conditions, and scales naturally with large datasets and token vocabularies. Although model families differ in whether they decode autoregressively, non-autoregressively, or through iterative refinement, Transformer-style attention remains a common architectural core.

E. DIFFUSION AND FLOW MATCHING

As the field moved beyond purely autoregressive decoding, diffusion and flow-based generation emerged as powerful alternatives. In diffusion models [5], training proceeds by progressively noising a clean sample along a forward trajectory and learning a denoiser that estimates the noise component at each step:

$$x_0 \sim P_{\text{data}},$$

$$x_t = \sqrt{\alpha_t}x_0 + \sqrt{1 - \alpha_t}\epsilon,$$

$$\epsilon \sim \mathcal{N}(0, I), \quad (5)$$

with a denoiser learning $\epsilon_\theta(x_t, t) \approx \epsilon$. At inference time, samples are generated by iteratively applying the trained denoiser to invert this trajectory from noise back to data. Flow matching [6] generalizes this idea by learning a continuous trajectory from noise to data:

$$\frac{d\phi_t}{dt} = v_t(\phi_t),$$

$$\phi_0 \sim \mathcal{N}(0, I),$$

$$\phi_1 \sim P_{\text{data}}. \quad (6)$$

In practice, flow matching has become especially attractive in TTS because it often trains more efficiently and provides a favorable balance between quality and inference speed. This explains why recent high-performing systems increasingly adopt flow-based designs.

The remainder of this survey compares systems along the taxonomy summarised in Table 2, which lists 15 representative systems side by side. The quantitative counterpart, Table 4, appears with the evaluation discussion in Section XIII.

IV. CODEC LANGUAGE MODELS REVOLUTION

With the necessary foundations in place, we can now turn to the first major inflection point of the modern era: the introduction of codec language models. This shift was important not merely because it improved objective metrics, but because it redefined the modeling problem itself. The quantitative view in Table 4 already covers training data scale, first-packet latency, and headline evaluation metrics (WER and speaker similarity) for an overlapping set of systems. Table 2 is the qualitative complement, covering architecture family, input representation, speaker conditioning, codec design, controllability, and open-source status. Together they provide a comparison across all eight axes without producing

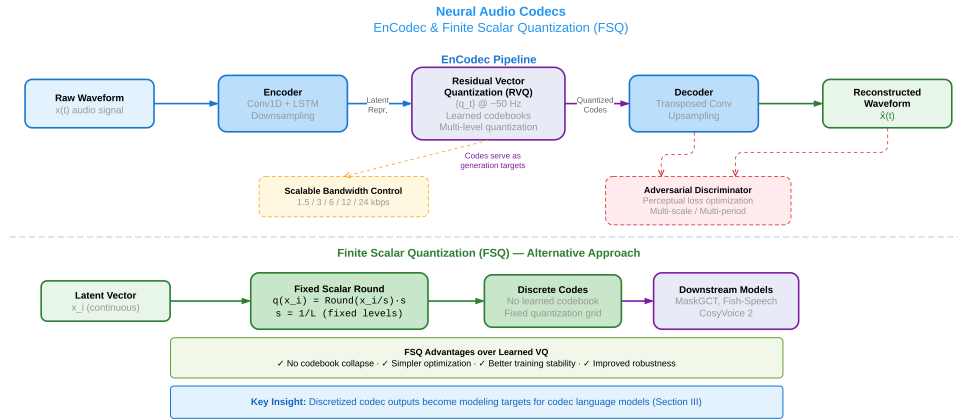


FIGURE 2. Neural audio codec foundations relevant to modern zero-shot voice cloning. The upper panel summarizes the EnCodec-style encoder–quantizer–decoder pipeline, while the lower panel contrasts it with finite scalar quantization as a simpler discretization strategy used by later TTS systems [3], [39].

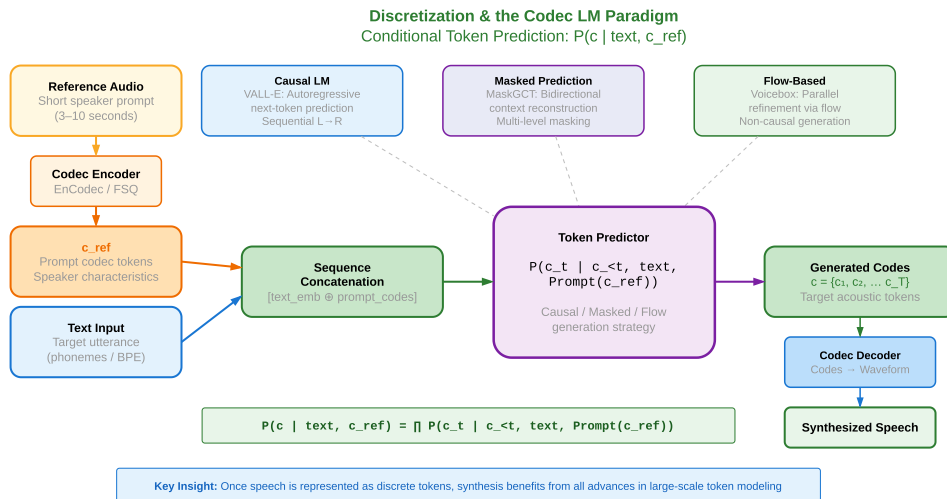


FIGURE 3. Discretization and the codec language model paradigm. A short reference utterance is converted into prompt code tokens, combined with text conditioning, and decoded through causal, masked-prediction, or flow-based generation before reconstruction as waveform speech [8], [9], [11].

a single table too wide to read on the page. This view makes the field’s architectural diversity explicit: systems with similar headline WER often differ fundamentally in generation strategy, conditioning interface, and codec design, and these differences, not the metric gaps, are what determine which system suits a given deployment.

A. VALL-E: THE MILESTONE

VALL-E is best understood as the point at which zero-shot TTS began to resemble language modeling over acoustic units. Instead of predicting waveform samples or spectrogram frames directly, it first represents speech using EnCodec tokens and then learns to generate the target acoustic-token sequence conditioned on text and a short reference utterance [3], [8]. This prompt-token interface is the reason the system became so influential: speaker identity is carried through the reference-code sequence rather than through speaker-specific retraining.

The reported results were strong for the period, including 5.9% WER and meaningful speaker-similarity gains after training on roughly 60K hours of speech [8]. More important than any single number, however, was the modeling template it introduced. The same design also exposed limitations that shaped later work: left-to-right decoding creates latency, prosodic control is mostly implicit, and the hierarchy among residual quantization levels is not fully exploited. Many later systems can be read as attempts to keep VALL-E’s codec-token interface while relaxing one or more of these constraints.

VALL-E 2 refined the family through improved sampling and training strategies, improving speaker similarity and narrowing the perceptual gap between generated and real speech [40]. Its role in the literature is therefore evolutionary rather than purely incremental: it strengthened the codec-LM template while leaving room for non-autoregressive and flow-based alternatives.

TABLE 2. Qualitative architectural taxonomy of zero-shot voice cloning systems. Architecture family: AR = autoregressive, NAR = non-autoregressive, Flow = flow matching, Diff = diffusion, Hybrid = combination (e.g., AR coarse + diffusion refinement). Input: char = character, phon = phoneme. Conditioning: prompt tokens = reference codec tokens in-context, embedding = fixed speaker vector, encoder = learned speaker encoder, text = natural-language style description. Codec/representation and open-source status as reported by the original works.

System	Arch. family	Input	Speaker cond.	Codec/repr.	Control features	Open
VALL-E [8]	AR	phon	prompt tokens	EnCodec RVQ	implicit prosody	No
VALL-E 2 [40]	AR	phon	prompt tokens	EnCodec RVQ	implicit prosody	No
Voicebox [9]	NAR (Flow)	phon	prompt (masked)	mel + flow	infilling, editing, non-causal	No
NaturalSpeech 3 [41]	NAR + Diff	phon	factorized embedding	factorized (FACodec)	content/prosody/timbre	Partial
MaskGCT [11]	NAR (masked)	phon	prompt tokens	VQ (semantic + acoustic)	duration control	Yes
F5-TTS [10]	Flow (NAR)	char	prompt (in-context)	mel + flow	none (duration-free)	Yes
CosyVoice 2 [13]	AR + Flow	text/phon	encoder + prompt	FSQ speech tokens	instruction, streaming	Yes
CosyVoice 3 [42]	AR + Flow	text/phon	encoder + prompt	supervised multi-task tok.	instruction, streaming	Yes
GLM-TTS [16]	AR	text	prompt tokens	RVQ tokens	preference-aligned (GRPO)	No
Qwen3-TTS [29]	AR	text	prompt tokens	RVQ (dual-rate)	streaming, low-latency	Yes
Seed-TTS [12]	Hybrid (AR + Diff)	text	prompt tokens	codec tokens	emotion, DPO/PPO align.	No
MiniMax-Speech [31]	Hybrid (Flow + VAE)	text	learnable encoder	Flow-VAE latent	identity/prosody/timbre	No
XTTS [43]	AR	char/phon	encoder	VQ-VAE + GPT	multilingual, streaming	Yes
YourTTS [7]	NAR (VITS-style)	phon	embedding	flow (VITS)	multilingual	Yes
Spark-TTS [26]	AR	text	decoupled tokens	pitch/energy/spectral tok.	attribute control	Yes

Notes: “Open” reflects availability of model weights or a reference implementation as reported by the authors; “Partial” indicates code or components released without full training pipeline or weights. Conditioning and codec descriptions summarize the primary mechanism; several systems combine multiple representations (e.g., semantic + acoustic streams). Categories are intended for comparative orientation and abstract away implementation-specific detail.

B. VOICEBOX: FLOW MATCHING AT SCALE

Voicebox broadened the codec-generation view by replacing strictly left-to-right prediction with a flow-matching formulation [9]. Instead of committing to one-token-at-a-time generation, the model learns to refine masked or noisy acoustic representations under text, speaker, and partial-audio conditioning. This makes the architecture naturally suited not only to full-utterance synthesis, but also to infilling, editing, and conditional transformation.

The significance of Voicebox is that it separated codec-based TTS from the assumption that generation must be causal. Parallel or non-causal refinement can reduce latency, while flexible conditioning makes it easier to incorporate prompts, style cues, and duration constraints. This opened a major branch of subsequent work in non-autoregressive and flow-based zero-shot synthesis.

C. HierSpeech++: SEMANTIC-ACOUSTIC BRIDGING

HierSpeech++ approached the problem from the perspective of representation separation. It uses WavLM-derived semantic representations for content and EnCodec-style acoustic representations for speaker and signal detail, then learns to bridge these streams during synthesis [3], [23], [44]. The strength of this design is conceptual clarity: content and speaker-related acoustics are separated before generation rather than expected to disentangle themselves implicitly. The cost is additional system complexity, but the approach strongly influenced later work on controllability, style separation, and robust cross-speaker synthesis.

D. Mega-TTS AND Mega-TTS 2

Mega-TTS and Mega-TTS 2 introduced more structure into prompt conditioning by separating global speaker identity, utterance-level variation, and fine acoustic detail into different representational roles [24], [25]. This design reduces

the burden on a monolithic end-to-end model to discover all useful decompositions by scale alone. In retrospect, the Mega-TTS line anticipated a broader trend: practical voice-cloning systems need organized speaker-conditioning spaces if they are to support personalization, controllability, and stable synthesis without extensive fine-tuning.

V. NON-AUTOREGRESSIVE, DIFFUSION, AND FLOW-MATCHING PARADIGMS

If codec language models established the new interface for speech generation, the next question was how best to generate those tokens. The answer diversified rapidly. Researchers increasingly moved beyond purely autoregressive decoding and began to exploit non-autoregressive alignment, iterative refinement, and flow-based generation.

A. F5-TTS: FLOW MATCHING WITHOUT DURATION MODELING

F5-TTS demonstrates that strong zero-shot synthesis can be obtained with a comparatively simple non-autoregressive flow-matching pipeline [10]. Its key design choice is to remove several modules common in earlier TTS systems: an explicit duration predictor, a separate text encoder, and a phoneme-level aligner. Instead, the character sequence $\mathbf{y}$ is padded to the target speech length T ,

$$\tilde{\mathbf{y}} = [\mathbf{y}_1, \dots, \mathbf{y}_{|\mathbf{y}|}, \langle \text{filler} \rangle, \dots, \langle \text{filler} \rangle] \in \mathbb{R}^T, \quad (7)$$

and alignment is learned jointly with acoustic generation. A ConvNeXt block refines the padded text representation before the Diffusion Transformer predicts flow velocities conditioned on the speaker prompt:

$$\frac{d\phi_t}{dt} = v_\theta(\phi_t, t, \text{ConvNeXt}(\tilde{\mathbf{y}}), \mathbf{x}_{\text{ref}}). \quad (8)$$

The authors also introduce Sway Sampling, which changes the allocation of flow-matching steps during inference

without requiring retraining [10]. The broader lesson is architectural: high quality and speaker similarity need not come from adding more alignment machinery; they can also emerge from simplifying the pipeline when the generative objective and conditioning interface are strong enough.

Related flow-matching systems. E2 TTS first showed that duration-free, alignment-free flow matching could implicitly learn text–audio alignment when the input is padded to the target length [45]. NaturalSpeech 3 combines factorized codec representations with latent diffusion, separating content, prosody, timbre, and acoustic detail into distinct factors [41]. FlashSpeech uses adversarial consistency training to accelerate flow-based inference, ZipVoice further explores fast flow matching for zero-shot synthesis, and MELLE shows that continuous mel-spectrogram prediction remains a viable alternative to discrete codec-token generation [46], [47], [48]. Together, these systems demonstrate that the codec-LM era is not architecturally uniform: flow matching, diffusion, masked prediction, and continuous acoustic modeling all remain active design choices.

B. MaskGCT: MASKED GENERATIVE CODEC TRANSFORMER

MaskGCT advances a different non-causal strategy based on masked prediction of codec tokens [11]. During training, both coarse and fine quantization levels are partially masked and reconstructed from surrounding context, text, and speaker conditioning:

$$m_{\text{RVQ}}^{(l)} \Rightarrow \hat{c}_{\text{hier}}^{(l)}, \quad l = 1, \dots, L. \quad (9)$$

This approach provides richer supervision than strictly causal next-token prediction because the model learns to reconstruct tokens from bidirectional context across multiple representational levels. The resulting gains in speaker similarity and intelligibility demonstrate the effectiveness of multi-level masked conditioning.

C. Seed-TTS: HYBRID AUTOREGRESSIVE AND DIFFUSION GENERATION

Seed-TTS is representative of a broader trend toward hybrid design [12]. Rather than committing fully to one generation strategy, it uses autoregression to model coarse structure and diffusion to restore fine acoustic detail. This combination is particularly effective because coarse tokens carry major timing and spectral information, while finer levels capture subtle details such as texture, prosody, and voice coloration. The model is further strengthened through self-distillation and preference-based post-training, making it influential both as a synthesis architecture and as an evaluation and training framework.

VI. MULTILINGUAL ZERO-SHOT TEXT-TO-SPEECH

The developments discussed so far largely concern the core synthesis problem, but multilinguality introduces a second layer of complexity. A multilingual zero-shot system must not only synthesize intelligible speech in many

languages, but also preserve a speaker’s identity when language, phonology, and prosodic conventions change. The central technical challenge is *cross-lingual speaker-identity preservation*: when a model trained primarily on language *A* is asked to synthesize the voice of a speaker enrolled in *A* but reading text in language *B*, several failure modes are common. Systems may produce *accent leakage*, where the speaker’s native accent inappropriately colors the target language; *foreign-accent artifacts*, where unfamiliar phonemes are produced with audibly non-native quality; or *timbre drift*, where the speaker’s identity gradually shifts toward speakers more frequent in the target-language training distribution. These failure modes are aggravated when phonological inventories differ substantially (e.g., tonal vs. non-tonal languages, or languages with retroflex consonants absent in the training distribution), and when low-resource languages are underrepresented in pretraining. Recent systems address these challenges with markedly different design choices. MiniMax-Speech uses a learnable speaker encoder to separate identity, prosody, and timbre, while Qwen3-TTS emphasizes large-scale multilingual pretraining and streaming-oriented architecture [29], [31]. Both approaches improve cross-lingual fidelity over earlier baselines, but the trade-off between explicit factorization and data-scale-driven generalization remains open.

A. YourTTS: THE MULTILINGUAL PIONEER

YourTTS [7] established the feasibility of multilingual zero-shot TTS by showing that a shared multilingual phonetic representation could be combined with speaker conditioning to generate unseen speakers across languages:

$$\mathbf{e}_{\text{phn}}^{\text{multi}} + \mathbf{e}_{\text{spk}} \rightarrow \mathbf{h}_{\text{ac}} \rightarrow \hat{\mathbf{x}}. \quad (10)$$

Although later systems would achieve better fidelity and broader coverage, YourTTS remains foundational because it demonstrated that multilinguality and zero-shot speaker transfer need not be separate research problems. An orthogonal direction explores language-agnostic phonological features [49] as input representations: instead of language-specific grapheme-to-phoneme modules, universal articulatory features (manner, place, voicing) enable more graceful degradation on unseen languages.

B. XTTS: OPEN-SOURCE MULTILINGUAL DEPLOYMENT

XTTS [43] extends multilingual zero-shot synthesis into a more deployment-oriented regime. Its significance is practical as much as technical: it offers an open-source, widely adopted, streaming-capable multilingual system with low reference-audio requirements. The model relies on language-agnostic speaker conditioning and supports a wide range of major languages, thereby making multilingual zero-shot synthesis accessible to both researchers and practitioners. A complementary open-source line of work is OpenVoice [50], with a subsequent V2 release adding native multilingual support; together these models combine multilingual zero-shot voice cloning with explicit style and

emotion control while remaining lightweight enough for community deployment. Together with XTTS, OpenVoice defines much of the practical baseline for open-source multilingual zero-shot TTS used in production today.

C. MiniMax-Speech: SCALING MULTILINGUAL SPEAKER MODELING

MiniMax-Speech [31] pushes multilingual synthesis further through a jointly trained speaker encoder and a Flow-VAE architecture. The learnable speaker encoder explicitly decomposes speaker-related information:

$$\mathbf{s}_{\text{identity}}, \mathbf{s}_{\text{prosody}}, \mathbf{s}_{\text{timbre}} = \text{SpeakerEncoder}(\mathbf{x}_{\text{ref}}), \quad (11)$$

and the Flow-VAE training objective is the evidence lower bound (ELBO), which combines reconstruction and latent prior matching and is *maximized* during training:

$$\begin{aligned} \mathcal{L}_{\text{ELBO}} = & \mathbb{E}_{q_{\phi}}[\log p_{\psi}(\mathbf{x}|\mathbf{z})] \\ & - D_{\text{KL}}(q_{\phi}(\mathbf{z}|\mathbf{x})\|p_{\theta}(\mathbf{z}|\text{cond})). \end{aligned} \quad (12)$$

The broader lesson from multilingual systems is that speaker identity can often be represented in a language-independent way, but only if the architecture and training data are sufficiently diverse and well balanced.

VII. DISENTANGLEMENT AND REPRESENTATION LEARNING

At this point, a recurring question becomes unavoidable: how should a model separate what is being said from who is speaking and how it is being said? This question underlies much of the recent literature on controllability and robustness. An idealized decomposition of speech may be written as

$$\text{Speech} = f(\mathbf{z}_{\text{content}}, \mathbf{z}_{\text{speaker}}, \mathbf{z}_{\text{style}}, \dots). \quad (13)$$

Although perfect disentanglement is unattainable in practice because these factors interact, partial disentanglement is both feasible and useful. HierSpeech++ pursues this goal through separate semantic and acoustic streams. StyleTTS 2 [15] approaches it through a style encoder, adversarial learning, and guidance from a frozen large speech language model:

$$\mathbf{z}_{\text{style}} = \text{StyleEncoder}(\mathbf{x}_{\text{ref}}), \quad (14)$$

with training objective

$$\begin{aligned} \mathcal{L}_{\text{style}} = & \mathcal{L}_{\text{MSE}} + \lambda_{\text{adv}} \mathcal{L}_{\text{adversarial}} \\ & + \lambda_{\text{lm}} \mathcal{L}_{\text{LM guidance}}. \end{aligned} \quad (15)$$

DisCo-Speech [28] adds explicit content and style codes and regularizes them through mutual information minimization:

$$\mathcal{L} = \mathcal{L}_{\text{recon}} + \beta \cdot I(\mathbf{c}_{\text{content}}; \mathbf{c}_{\text{style}}), \quad (16)$$

so that minimizing $\mathcal{L}$ encourages low mutual information between the content and style codes. Vevo [51] complements these approaches with self-supervised contrastive learning, showing that useful controllable structure can be learned even without explicit emotion labels. Taken together, these systems

reflect a broader shift. Earlier large models often relied on scale to learn useful factorization implicitly. More recent systems, especially those targeting expressive or interactive applications, increasingly prefer explicit disentanglement to gain interpretability and control.

VIII. STREAMING AND REAL-TIME SYNTHESIS

As voice cloning moves from offline generation to interactive systems, latency becomes a primary design constraint rather than a secondary optimization target. This shift changes the architecture of the problem itself, because streaming systems cannot assume access to complete future context.

A. CosyVoice 2

CosyVoice 2 [13] exemplifies a production-oriented streaming design built around compact model size (0.5B parameters), chunked decoding, FSQ-style quantization, and small look-ahead buffers. Its first-packet latency of approximately 150 ms enables conversational deployment, while its training on 200K hours of data supports broad coverage.

CosyVoice 3 [42] scales this line of work substantially, increasing training data to one million hours across 9 languages and 18 Chinese dialects, and growing model capacity from 0.5B to 1.5B parameters. A key architectural contribution is a supervised multi-task speech tokenizer that improves prosody naturalness over self-supervised alternatives, along with a differentiable reward model for post-training that further refines speaker similarity and naturalness through reinforcement-based fine-tuning.

B. Qwen3-TTS

Qwen3-TTS [29] advances streaming further through a dual-track architecture. A coarse decoder (Track A, 25 Hz) emits tokens for lower RVQ levels immediately as text arrives, producing audible output within approximately 97 ms. A fine decoder (Track B, 12 Hz) runs asynchronously to fill in higher RVQ levels with spectral detail, conditioned on Track A output with a small lookahead. A lightweight vocoder merges both tracks into a single waveform in real time. This design exploits the information asymmetry in RVQ: coarse levels carry phonemic and pitch information sufficient for intelligible speech, while fine levels add texture, breathiness, and speaker-specific coloration.

C. Fish-Speech AND Fish-Speech S2

Fish-Speech [14] combines large-scale multilingual training (720K hours, primarily Mandarin and English) with a dual autoregressive architecture and grouped finite scalar quantization (GFSQ). GFSQ partitions the latent vector into G groups, each independently scalar-quantized with L levels, avoiding exponential codebook growth while preserving gradient flow. Fish-Speech S2 [30] extends this with improved streaming and multilingual performance. The trade-off across these systems is straightforward but important: streaming reduces access to global context and therefore risks lower prosodic quality or weaker coherence.

Successful systems counterbalance this loss through hierarchical token generation, limited look-ahead, efficient model sizing, and asynchronous refinement. A key methodological issue remains the lack of standardization in latency reporting, since some papers measure only model inference while others include tokenization, vocoding, buffering, and application-level overhead.

IX. REINFORCEMENT LEARNING AND PREFERENCE ALIGNMENT

The move toward preference alignment reflects another maturation of the field. Once synthesis quality became reasonably strong, likelihood-based objectives alone no longer captured the criteria users actually care about most. A system may produce intelligible speech and still sound unnatural, emotionally inappropriate, or weakly matched to the target speaker.

A. MOTIVATION

Conventional training minimizes supervised reconstruction-like losses:

$$\mathcal{L}_{\text{supervised}} = \mathbb{E}_{(\mathbf{x}, \mathbf{y})} [\|\text{model}(\mathbf{y}) - \mathbf{x}\|^2]. \quad (17)$$

Such objectives do not directly optimize perceived quality. This is why preference-based post-training has become increasingly attractive.

B. GLM-TTS: GROUP RELATIVE POLICY OPTIMIZATION

GLM-TTS [16] applies Group Relative Policy Optimization (GRPO) to zero-shot speech generation, rather than standard Direct Preference Optimization (DPO) or Proximal Policy Optimization (PPO). GRPO generates a group of G candidate outputs per input, scores them with a reward model, and computes group-relative advantages for policy updates. The full objective combines a clipped surrogate term with a KL penalty against the reference policy:

$$\begin{aligned} \mathcal{L}_{\text{GRPO}} = & -\frac{1}{G} \sum_{i=1}^G \min \left(\frac{\pi_{\theta}(y_i|\mathbf{x})}{\pi_{\text{ref}}(y_i|\mathbf{x})} A_i, \right. \\ & \left. \text{clip} \left(\frac{\pi_{\theta}(y_i|\mathbf{x})}{\pi_{\text{ref}}(y_i|\mathbf{x})}, 1-\epsilon, 1+\epsilon \right) A_i \right) \\ & + \beta D_{\text{KL}}(\pi_{\theta} \parallel \pi_{\text{ref}}), \end{aligned} \quad (18)$$

where $A_i = (r_i - \bar{r})/\sigma_r$ is the advantage of sample i relative to the group mean $\bar{r}$ and standard deviation σ_r , and β controls the strength of the KL regularizer that limits drift from the supervised reference policy. This group-relative normalization avoids the need for a separately trained value network (as in PPO) or paired preference data (as in DPO), instead deriving reward baselines directly from the group statistics. GLM-TTS reports CER of 1.03% on Seed-TTS-eval-zh and speaker similarity of 0.764 on the same benchmark, demonstrating that GRPO-based alignment meaningfully improves perceived quality beyond supervised training.

C. Seed-TTS AND Spark-TTS

Seed-TTS [12] extends preference alignment through a hybrid generation pipeline (autoregressive coarse codes followed by diffusion refinement) with DPO and PPO post-training. The DPO objective operates on paired outputs:

$$\begin{aligned} \mathcal{L}_{\text{DPO}} = & -\log \sigma \left(\beta \left[\log \frac{\pi_{\theta}(\mathbf{y}^+|\mathbf{x})}{\pi_{\text{ref}}(\mathbf{y}^+|\mathbf{x})} \right. \right. \\ & \left. \left. - \log \frac{\pi_{\theta}(\mathbf{y}^-|\mathbf{x})}{\pi_{\text{ref}}(\mathbf{y}^-|\mathbf{x})} \right] \right), \end{aligned} \quad (19)$$

where $\mathbf{y}^+$ is preferred and $\mathbf{y}^-$ is dispreferred. Combined with self-distillation for model compression, this pipeline achieves CMOS scores approaching human parity. Spark-TTS [26] combines alignment with efficient decoupled token design, representing acoustic features through separate pitch, energy, and spectral tokens rather than standard RVQ levels. This token factorization reduces sequence length (50 tokens/sec vs. typical 150–200 for RVQ-based systems) while maintaining quality (UTMOS 4.35).

D. EMERGING ALIGNMENT METHODS

Recent work has expanded the alignment toolkit beyond DPO and PPO. *Fine-grained Preference Optimization (FPO)* [52] addresses a key limitation of utterance-level DPO by introducing token- or segment-level preference labels: rather than uniformly optimizing the entire utterance, FPO targets only the localized regions in which generation issues actually occur (e.g., specific mispronounced words or unnatural prosody). This selective training strategy reduces the bad-case ratio and improves data efficiency relative to coarse-grained baselines. *ARDM-DPO* [53] adapts preference alignment to autoregressive diffusion models by operating at the trajectory level, optimizing the cumulative denoising policy to prefer trajectories leading to human-preferred outputs. This is significant because standard DPO assumes single-step generation and cannot be directly applied to iterative diffusion processes.

A critical gap in current alignment work concerns reward-model overoptimization [54]: because the reward model is an imperfect proxy for true human preferences, optimizing its value too aggressively can hinder ground-truth performance, in accordance with Goodhart's law. In the TTS setting this manifests as a divergence between automatic preference scores and listener judgments as training progresses, producing models that achieve high automatic scores while sounding subtly unnatural. A concrete TTS-domain instance involves models optimized against neural MOS-prediction rewards such as UTMOS [55]: aggressive optimization can drive predicted-MOS scores upward while listener panels rate the resulting samples no better, and sometimes worse, than the supervised baseline, because the reward model rewards spectral and prosodic patterns it was trained to prefer rather than perceptual naturalness per se. This motivates periodically refreshed reward models, diverse annotator

pools, and KL-regularized objectives that limit policy drift from the supervised reference.

The importance of this entire line of work is that it shifts the optimization target from reconstruction fidelity toward human satisfaction. The main obstacles are practical: preference data are expensive, reward models can drift, and alignment quality depends strongly on annotation diversity and protocol design.

X. CONTROLLABILITY AND STYLISTIC FLEXIBILITY

After quality and latency, controllability has become one of the defining themes of recent research. Users increasingly expect more than faithful cloning; they want the ability to control emotion, timing, emphasis, speaking style, and conversational persona.

IndexTTS2 [17] addresses this need through explicit emotion–timbre disentanglement. A text-to-semantic autoregressive stage predicts semantic tokens conditioned on soft emotional instructions, and a semantic-to-acoustic flow-matching stage produces acoustic features conditioned on the semantic content c_{sem} together with two separate emotion-related streams: a suppression stream that removes unintended emotional content from the reference c_{ref} , and a target stream that injects the desired emotion c_{target} . Schematically:

$$z_t = \text{FlowMatch}(z_0, c_{\text{sem}}, \text{Suppress}(c_{\text{ref}}, \text{emotion}), c_{\text{target}}), \quad (20)$$

where the three conditioning arguments are passed as separate streams to the flow-matching network rather than combined arithmetically.

EmoSphere++ [56] instead models emotion on a hypersphere, using geodesic interpolation (slerp) between emotion vectors to produce smoother affective transitions than Euclidean baselines. Parler-TTS [57] takes a more user-facing route by replacing reference-based speaker control with natural-language descriptions of the desired voice. ControlSpeech [58] uses mixture density networks for fine-grained style interpolation through mixture component selection. These approaches differ in mechanism but share a common goal: to make zero-shot speech generation more interpretable and more useful in creative and interactive settings. They also reveal a consistent trade-off. As controllability becomes more explicit and fine-grained, raw intelligibility metrics may not always improve correspondingly. The challenge is therefore to design systems that preserve cloning fidelity while exposing meaningful control.

Industry Adoption. Commercial platforms have integrated controllability features at scale. ElevenLabs V3 [59] enables voice cloning with fine-grained control via tagging systems and contextual prompting, and OpenAI’s GPT-4o real-time voice mode [60] demonstrates that large multimodal models can support voice-conditioned interactive synthesis, though its internal architecture has not been publicly described. ChatTTS [61] provides an open-source conversational TTS

system emphasizing natural prosody and filled pauses, and has seen rapid community adoption. These commercial and open-source systems collectively demonstrate that controllable zero-shot synthesis is transitioning from a research capability to a deployment expectation.

XI. CONVERSATIONAL AND FULL-DUPLEX VOICE SYNTHESIS

The next extension of controllable synthesis is conversational deployment. A full-duplex voice agent must do more than generate speech quickly; it must coordinate timing, interruption, listener feedback, and acoustic interaction with the environment. Turn-taking models increasingly rely on semantic and prosodic prediction rather than simple silence thresholds. Echo cancellation must handle nonlinear and speaker-dependent acoustic feedback, especially when the synthesized voice is played over external speakers. Backchannel generation introduces another layer of complexity, as short listener responses must be inserted with appropriate timing and prosody. Several concrete systems illustrate the design space. Moshi [62] treats the dialogue agent as a single end-to-end speech-text model with a continuously running input and output stream, enabling natural overlapping speech and short-latency interruption. OpenAI’s GPT-4o real-time voice mode [60] demonstrates a comparable interactive regime within a large multimodal model, although its internal architecture has not been publicly disclosed. Hume AI’s Empathic Voice Interface [63] adds explicit affect prediction on top of the conversational pipeline, conditioning synthesis on inferred listener emotional state. PersonaPlex [18], discussed further in Section XII, similarly targets the full-duplex regime with explicit voice and role control. Benchmarks such as FullDuplexBench [64] show that turn-taking, interruption, and backchannel generation matter significantly for perceived naturalness, even when the underlying synthesis quality is already high.

XII. MULTIMODAL INTEGRATION: SPEECH, VISION, AND TEXT

Recent systems also expand beyond audio-only conditioning. Video-conditioned synthesis, such as that explored in PersonaPlex [18], uses facial or identity embeddings to supplement noisy or limited reference audio:

$$\hat{\mathbf{x}} = \text{ZS-VC}_{\text{joint}}(\text{text}, \mathbf{x}_{\text{ref}}, \mathbf{e}_{\text{face}}). \quad (21)$$

MM-Sonate [19] integrates dialogue context, sentiment, and speaker relationship information. Context-TTS [65] incorporates conversational history and emotional context to improve synthesis coherence in dialogue. Although multimodal gains are not always large in absolute terms, they become increasingly valuable in noisy, conversational, or context-rich settings.

XIII. DATASETS, BENCHMARKS, AND EVALUATION

Evaluation remains one of the least standardized aspects of the field, and this has important consequences for scientific

comparison. Reported improvements are only meaningful when systems are tested on comparable data, with comparable metrics, under contamination-free protocols. Table 3 maps each common metric to the quality dimension it captures and to the way it tends to diverge from listener judgment.

A. BENCHMARKS AND EVALUATION FRAMEWORKS

Seed-TTS-eval [12] has emerged as a widely adopted evaluation benchmark for recent zero-shot voice cloning systems, used in major system papers published including MaskGCT, F5-TTS, Seed-TTS, CosyVoice 2, MiniMax-Speech, and Spark-TTS. It comprises 1,000 utterances across 200 speakers with balanced gender and multilingual coverage (English, Mandarin, Spanish, French, German, Japanese), and provides standardized scripts for WER and speaker similarity computation. Its adoption has driven rapid progress: systems achieving below 3.5% WER are now baseline expectations. However, adoption is not universal: GLM-TTS reports on Seed-TTS-eval-zh (Chinese subset), and earlier systems like YourTTS report on VCTK-based benchmarks, limiting direct comparability. **ClonEval** [27] provides a complementary black-box evaluation protocol specifically designed for voice cloning. It uses WavLM-based speaker similarity [44] as the primary identity metric and enforces a standardized automated pipeline for reference selection, synthesis, and evaluation. Key limitations include its English-only focus and the absence of human MOS scores, which means it may miss perceptual quality dimensions that automatic metrics do not capture. Other evaluation resources include InstructTTS-eval [66] for instruction-following capabilities, PESQ [67] and STOI [68] for intelligibility assessment, UTMOS [55] for predicted MOS estimation, and LiveBench [69] for contamination-resistant evaluation through dynamic test set updates. **Standardization Recommendations.** We recommend that future work adopt the following evaluation protocol for improved cross-paper comparability: (1) **ASR backend:** Whisper-large-v3 [70] for WER computation, with explicit model version reported; (2) **Speaker similarity extractor:** WavLM-TDNN [44] or ECAPA-TDNN [71], reporting cosine similarity on a consistent enrollment/test protocol; (3) **Evaluation set:** Seed-TTS-eval as the primary benchmark, with VCTK and LibriSpeech test-clean as secondary cross-dataset validation; (4) **Confidence intervals:** 95% bootstrap confidence intervals for all metrics; (5) **Peer-review status:** clearly mark whether results are from peer-reviewed venues, preprints, or technical reports.

B. METRIC RELIABILITY AND THE AUTOMATIC-HUMAN GAP

Standardizing benchmarks (Section XIII-A) addresses what systems are tested on, but a second and equally consequential problem is what each metric actually measures. Zero-shot voice cloning is evaluated along at least four partly independent quality dimensions, namely intelligibility,

speaker similarity, naturalness, and prosody or expressiveness. No single metric captures all of them. Reporting a low WER alongside a high speaker-similarity score can create an impression of overall quality even when prosody is flat or affect is inappropriate, because neither metric is sensitive to those dimensions. Table 3 organizes the commonly used metrics by the dimension they primarily target, whether they are automatic or human, and their principal failure modes.

Two consequences follow. First, automatic metrics diverge from human judgment in predictable ways. ASR-based WER saturates once speech is intelligible and thereafter says nothing about whether it sounds human; embedding-based similarity can be inflated by train/test speaker overlap (Section XIII-D) and is demographically uneven; and learned MOS predictors such as UTMOS can be optimized against directly, so that predicted scores rise while listener ratings stagnate or fall. This is an instance of the reward-overoptimization effect discussed in the alignment literature [54]. Second, human evaluation remains essential precisely for the dimensions automatic metrics handle worst: fine-grained prosody, emotional appropriateness, and comparative naturalness near human parity, where CMOS differences of tenths of a point are perceptually real but below the resolution of current predictors. We therefore recommend that system papers report at least one metric per quality dimension, treat automatic naturalness predictors as screening tools rather than endpoints, and accompany near-parity claims with human CMOS/SMOS under a described protocol (rater count, instructions, and listening conditions).

A broader quality taxonomy also includes emotion, accent preservation, robustness, and safety. In this survey these are handled in adjacent sections rather than as separate speech-quality dimensions. Emotion is treated in Section X on controllability, where affect is a system output rather than a scalar to score. Accent preservation is treated in Section VI on cross-lingual identity preservation and is measured through cross-lingual speaker similarity rather than a separate metric. Robustness to demographic variation is treated in Section XIV on fairness and demographic auditing. Safety, including anti-spoofing, watermarking, and deepfake detection, is treated in Section XIV-B and Section XIV-A and follows a distinct EER-based evaluation regime rather than a naturalness metric.

Several broad trends are visible. Error rates have improved substantially since early codec-language-model systems, but later gains appear smaller than the initial jump from early baselines. Data scaling helps, though not uniformly; model architecture and codec design often matter at least as much as raw training hours. Streaming systems remain especially hard to compare because quality and latency are traded off differently across implementations.

C. TRAINING CORPORA

Common training corpora include Emilia [74], LibriSpeech [75], CommonVoice [76], GigaSpeech [77],

TABLE 3. Taxonomy of evaluation metrics for zero-shot voice cloning. Organized by quality dimension. “Type” distinguishes automatic (A) from human (H) protocols. Each metric is most informative for the dimension it primarily targets and is weak or uninformative for the others; over-reliance on any single metric is a recurring source of misleading comparison.

Metric	Dimension	Type	What it captures	Principal limitation
WER / CER (via ASR, e.g. Whisper [70])	Intelligibility	A	Whether synthesized content is correctly recognized	Blind to naturalness, similarity, prosody; sensitive to ASR backend and language
Speaker sim. (WavLM/ECAPA cosine) [44], [71]	Speaker similarity	A	Identity match to reference in an embedding space	Trained on binary-gender/limited demographics; inflated by contamination
PESQ [67]	Signal fidelity	A	Perceptual codec-style signal quality	Designed for enhancement/telephony, not generative naturalness
STOI [68]	Masked intelligibility	A	Short-time intelligibility under noise or masking	Not sensitive to voice cloning fidelity
Mel-cepstral distortion [72]	Signal / spectral	A	Spectral distance to a reference utterance	Requires aligned reference; penalizes valid prosodic variation
UTMOS [55]	Naturalness (predicted)	A	Learned proxy for mean opinion score	Reward-hackable; diverges from listeners under aggressive optimization
MOS / CMOS	Naturalness	H	Human-rated naturalness (absolute / comparative)	Costly; sensitive to rater pool, instructions, listening conditions
SMOS	Speaker similarity	H	Human-rated identity match	Rater-dependent; hard to separate from naturalness
ITU-T P.808 [73]	Naturalness / quality	H	Crowdsourced MOS with reliability controls	Still requires careful screening; crowd conditions vary
Preference / A-B tests	Overall / prosody	H	Relative preference between systems	Non-transitive; masks which dimension drove the choice

TABLE 4. Quantitative comparison of zero-shot voice cloning systems. WER and speaker similarity (Spkr Sim) values are reported as published in the original papers; ASR backends (e.g., Whisper-large-v3 [70], HuBERT-large, Paraformer-zh) and speaker-similarity extractors (e.g., WavLM-TDNN [44], ECAPA-TDNN) vary across systems and are not normalized in this table. See Section XIII-A for our standardization recommendations. “N/R” = not reported. † indicates a different metric than standard WER (see notes). ‡ indicates preprint without peer review. Table entries are drawn from original publications and may not be directly comparable across different evaluation sets and ASR backends.

System	WER↓	Spkr Sim↑	Params	Train Data	Eval Set	Year	Streaming	Latency
VALL-E [8]	5.9%	0.580	450M	60K hrs	LibriSpeech	2023	No	N/R
VALL-E 2 [40]	3.6%	0.643	450M	60K hrs	LibriSpeech	2024	No	N/R
NaturalSpeech 3 [41]	1.81%	0.67	500M	200K hrs	LibriSpeech	2024	No	N/R
MaskGCT [11]	2.634%	0.751	385M	100K hrs	Seed-TTS-eval	2024	No	N/R
F5-TTS [10]	2.42%	0.66	336M	100K hrs	Seed-TTS-eval	2024	No	N/R
CosyVoice 2 [13]†	3.2%	0.71	0.5B	200K hrs	Seed-TTS-eval	2024	Yes	150 ms
CosyVoice 3 [42]‡	2.1%	0.76	1.5B	1M hrs	Seed-TTS-eval	2025	Yes	N/R
GLM-TTS [16]‡	1.03% CER†	0.764	1.3B	100K hrs	Seed-TTS-eval-zh	2025	No	N/R
Qwen3-TTS [29]‡	1.24%	0.78	1.7B	>5M hrs	Seed-TTS-eval	2026	Yes	97 ms
Seed-TTS [12]	3.1%	0.715	0.7B	150K hrs	Seed-TTS-eval	2024	No	N/R
MiniMax-Speech [31]‡	2.8%	0.74	0.8B	180K hrs	Seed-TTS-eval	2025	Yes	120 ms
XTTS [43]	3.5%	0.69	467M (GPT bb.)	130K hrs	VCTK	2024	Yes	<200 ms
YourTTS [7]	4.2%	0.65	350M	1.2K hrs	VCTK	2022	No	N/R
Spark-TTS [26]‡	2.7%	0.75	0.6B	120K hrs	Seed-TTS-eval	2025	Yes	140 ms

Notes: GLM-TTS reports CER (Character Error Rate) on the Chinese subset of Seed-TTS-eval (Seed-TTS-eval-zh) using GRPO-based alignment; direct comparison with WER-based entries requires caution due to metric and language differences. NaturalSpeech 3 results are from the LibriSpeech test-clean split. YourTTS was trained on substantially less data (1.2K hours) than later systems. Streaming latency refers to first-packet latency where reported; measurement methodology varies across papers (see Section VIII for discussion). XTTS parameter count reflects the GPT-2 backbone only; the full system (including codec decoder and vocoder) is larger. Rows combine results from different benchmarks, languages, ASR backends, and speaker-similarity extractors; values should be interpreted as paper-reported indicators rather than strictly standardized head-to-head comparisons. Human MOS is not uniformly reported by original papers and cannot be reconstructed post hoc, so it is omitted from this table; per-system recommendations for human evaluation reporting are given in Section XIII-B. All values reflect numbers as published; readers should consult Section XIII-D for a discussion of contamination-related caveats that may affect headline metrics on LibriSpeech-derived evaluation sets in particular.

WenetSpeech [78], LibriTTS [79], VCTK [80], MLS [81], and LibriLight [82]. For Indic and lower-resource settings, datasets such as IndicVoices [83] and SysPIN [84] are increasingly important.

D. DATA CONTAMINATION AND LEAKAGE

A persistent evaluation risk is contamination between training and test sets. If the same speaker or utterance appears across both, reported speaker similarity and intelligibility

can be inflated. Recent work on benchmark contamination in LLM-based speech systems [85] demonstrates that substantial portions of LibriSpeech and Common Voice evaluation sentences appear in widely used LLM pretraining corpora, biasing reported metrics toward optimistic estimates. Although that audit targets ASR rather than zero-shot voice cloning specifically, the same underlying risk applies to zero-shot TTS systems trained or evaluated on overlapping corpora: a model can appear to generalize to “unseen”

speakers or utterances that are in fact present in some form during training. Contamination can operate at two levels: speaker-level overlap (same speaker in train and test with different utterances), which primarily inflates similarity scores, and utterance-level overlap (identical audio or transcripts), which affects both WER and similarity. Public contamination audits for individual zero-shot voice cloning systems remain rare, representing an important gap. Stronger evaluation practice therefore requires published speaker split definitions, multilingual overlap analysis, cross-dataset testing, and transparent reporting of overlap at both the speaker and utterance levels. Benchmarks such as Seed-TTS-eval move in this direction, but broader field-wide standardization is still needed.

XIV. SECURITY, FAIRNESS, AND ETHICAL CONSIDERATIONS

As synthesis quality improves, the risks associated with impersonation, fraud, and non-consensual cloning become more acute. This is no longer a peripheral concern but a central deployment constraint. A consolidated pre-deployment checklist follows the technical treatment in Table 5.

A. WATERMARKING AND PROVENANCE

VoiceMark [20] and AudioSeal [86] illustrate two watermarking strategies: one operating on codec representations and the other at the waveform level. Both aim to support downstream detection and provenance tracking, though both also face adversarial removal risks. These systems are important because they provide a technical basis for post-hoc verification, but they cannot prevent misuse alone. Legal and regulatory frameworks, including the EU AI Act [87], introduce transparency obligations for certain AI-generated or AI-manipulated content, increasing the importance of provenance and disclosure for synthetic speech.

B. VOICE AUTHENTICATION AND ANTI-SPOOFING

Voice-based authentication systems are under growing pressure from high-quality cloned speech. The ASVspoof challenge series [88] provides the primary benchmark for evaluating spoofing countermeasures. ASVspoof introduced evaluation tracks for both logical access (synthesized speech) and deepfake detection scenarios, with equal error rate (EER) as the primary metric. State-of-the-art countermeasures achieve EER below 1% on known attack types but degrade significantly (EER > 15%) against unseen synthesis methods, highlighting the need for attack-agnostic detection. As ZS-VC systems improve, spoofing countermeasures must be updated continuously; the community would benefit from rolling benchmarks analogous to Seed-TTS-eval but focused on detection rather than synthesis quality.

C. MACHINE UNLEARNING AND CONSENT

Machine unlearning and consent frameworks are becoming increasingly important as regulatory environments evolve. Speaker unlearning [89] provides mechanisms to remove a

speaker's voice from trained models without full retraining, while consent frameworks must address explicit permission, use-case limitation, transparency, and equitable benefit sharing.

D. DEMOGRAPHIC BIAS AND FAIRNESS

Fairness remains an equally significant concern. Current ZS-VC systems can exhibit measurable demographic bias, especially for accents, languages, and speaker groups that are underrepresented in training and evaluation corpora [2]. Recent work on gender-ambiguous voice generation [90] highlights an underexplored dimension: many applications benefit from voices that do not strongly signal a particular gender. Generating such voices is technically challenging because most speaker verification systems, including ECAPA-TDNN [71] and WavLM [44], are trained on binary-gendered speaker labels, making evaluation of gender-ambiguous synthesis difficult with standard metrics. Evaluation itself can also be biased because speaker similarity tools are frequently trained on demographic assumptions that do not transfer evenly across populations. As a result, progress in zero-shot voice cloning must be accompanied by stronger demographic auditing and more diverse data collection.

E. DATASET LICENSING AND SPEAKER RIGHTS

The technical safeguards above address misuse of deployed systems, but a distinct and often overlooked set of concerns arises at the training-data stage: who owns the voices a model learns from, and under what terms. Large zero-shot systems are trained on tens to hundreds of thousands of hours of speech, and the provenance of that audio determines both the legality of the resulting model and the rights of the speakers whose voices it can reproduce. Common corpora differ substantially in their licensing posture. Read-speech corpora derived from public-domain audiobooks (e.g., LibriSpeech and LibriTTS, from LibriVox) carry permissive terms, and crowd-donated corpora such as Common Voice release the audio under CC0 while surrounding platform terms of service still apply. Many large-scale training sets are assembled by scraping in-the-wild audio whose speakers never consented to voice-model training, and web-scale corpora such as Emilia [74] frequently mix licenses without per-utterance provenance. Because a zero-shot model can reproduce a recognizable voice from only seconds of reference audio, a speaker's presence in the training data can translate directly into the capacity to synthesize their identity, which is a risk that ordinary content licenses were not written to address.

Consent for voice data should therefore be treated as distinct from consent for ordinary content: it needs to be informed (the speaker understands that a reproducible voice model may result), purpose-limited, and ideally revocable, which connects speaker rights to the machine-unlearning mechanisms discussed above. A growing body of regulation and case law is beginning to codify these expectations.

TABLE 5. Pre-deployment recommendations checklist for zero-shot voice cloning systems. The checklist consolidates the safety, fairness, and licensing considerations of Section XIV into concrete actions, the technical or governance mechanism that supports each, and the primary risk it mitigates. It is intended as a practitioner-facing baseline rather than an exhaustive compliance standard.

Action	Mechanism / how	Risk mitigated	Status
Verify speaker consent	Informed, purpose-limited, revocable consent for both training data and target voices; record provenance and licenses (this section)	Non-consensual cloning; training-data rights violations	Required
Embed provenance watermark	Codec- or waveform-level watermarking (VoiceMark [20], AudioSeal [86]) applied to all generated audio	Undetectable synthetic media; loss of provenance	Required
Disclose synthetic origin	Transparency labeling per EU AI Act obligations [87]; user-facing “AI-generated” notice	Deception; regulatory non-compliance	Required
Integrate spoof detection	Anti-spoofing countermeasures benchmarked on ASVspoof [88]; monitor for unseen-attack degradation	Voice-authentication bypass; fraud	Required
Conduct bias audit	Evaluate WER and speaker similarity across accents, languages, and demographic groups; report disaggregated results	Demographic performance disparities	Required
Enable speaker removal	Machine-unlearning / speaker-unlearning support [89] for opt-out and deletion requests	Inability to honor revocation; GDPR [91] erasure exposure	Required
Rate-limit and gate cloning	Identity verification and usage limits on cloning of arbitrary target voices	Impersonation at scale; harassment	Recommended
Log and monitor usage	Auditable generation logs with retention policy	Abuse detection; incident response	Recommended

The EU AI Act [87] imposes transparency obligations on providers of systems that generate or manipulate audio, requiring that synthetic media be disclosed as such; data-protection regimes such as the GDPR [91] treat voiceprints as biometric personal data subject to consent and erasure rights; and jurisdiction-specific statutes, for example Tennessee’s ELVIS Act [92] extending right-of-publicity protection explicitly to voice, signal a shift toward recognizing voice identity as a protected attribute. Practitioners assembling training corpora should document data sources and licenses, obtain and record consent where speakers are identifiable, honor opt-out and deletion requests, and avoid redistributing cloned voices of identifiable individuals without permission.

This subsection consolidates six operational concerns: consent (Section XIV-C and this subsection), watermarking (Section XIV-A), speaker verification and anti-spoofing (Section XIV-B), deepfake detection (Section XIV-B and Section XV for cross-modal detection), dataset licensing (this subsection), and demographic bias (Section XIV-D). The checklist in Table 5 compresses these into concrete pre-deployment actions.

XV. FUTURE DIRECTIONS AND OPEN CHALLENGES

Looking ahead, several research directions appear especially consequential. **Unified speech-text foundation models** may improve instruction following and cross-lingual transfer by jointly modeling text and speech within a single architecture. Early results from multimodal models suggest potential relative WER gains, although scaling to multilingual settings remains open. **Real-time emotion trajectory control** could make expressive synthesis more natural within long utterances rather than only at the utterance level. Current systems control emotion globally per utterance; advancing to smooth within-utterance emotion evolution (e.g., neutral to emphatic

to calm) requires differentiable emotion trajectories that respect phonetic boundaries. **Cross-modal deepfake detection** will likely become increasingly necessary as multimodal synthetic media matures. Audio-video desynchronization, semantic inconsistency between modalities, and characteristic synthesis artifacts provide complementary detection signals. **On-device zero-shot synthesis** will matter more as privacy and latency constraints push speech generation to mobile and embedded platforms. Recent industry-facing on-device TTS systems such as NeuTTS Air [93] suggest that real-time voice cloning on mobile-class hardware is becoming increasingly feasible, although published parameter and deployment details remain limited. Typical deployment targets discussed in the on-device speech and on-device LLM communities include memory budgets on the order of a few hundred megabytes for quantized models, CPU-only real-time factor requirements below 1.0 on modern ARM processors, modest sustained power budgets, and the need for standardized first-packet latency measurement at the application level rather than model-level forward-pass time alone. **Synthetic data for downstream tasks.** Beyond direct speech generation, zero-shot TTS has significant value as a data augmentation tool. Recent work [94] shows that the high voice diversity afforded by zero-shot voice cloning makes synthetic ASR training data substantially more effective than synthetic data produced by conventional single-speaker TTS, with low-variability augmentation saturating quickly while high-variability synthetic speech continues to provide gains as more data is added. This suggests that as ZS-VC quality improves, it can help address data scarcity across speech processing more broadly. **Personalized voice agents** that adapt to user preferences over time will connect controllability with long-term interaction, and may benefit from continual alignment methods that update speaker models from implicit user feedback.

XVI. CONCLUSION

Zero-shot voice cloning has progressed from an early proof of concept to a mature and rapidly diversifying area of speech generation research. The most important development over the past few years has not simply been a reduction in error rates, but a broader transformation in how the field defines success. Early work established feasibility by showing that short prompts could support recognizably speaker-consistent synthesis. Later work demonstrated that discrete codec representations, large-scale sequence modeling, and flow-based generation could raise quality significantly. Most recent work has extended the problem again, emphasizing streaming capability, controllability, human preference alignment, multimodal conditioning, and secure deployment. Three broad conclusions emerge from the literature reviewed here. First, the codec language model perspective remains one of the field's defining conceptual advances because it connects speech synthesis to the broader ecosystem of token-based generative modeling. Second, the strongest recent systems succeed not by relying on a single universal recipe, but by carefully coordinating representation learning, conditioning design, generation strategy, and deployment constraints. Third, the field's next inflection point will be defined by deployment-grade trustworthiness encompassing robustness, interactivity, multilingual fairness, and provenance rather than by incremental gains on existing quality metrics. Accordingly, the principal open challenges are now clearer. Multilingual performance remains uneven, especially for low-resource languages and accents. Explicit control over emotion, timing, and non-speech vocalization is still incomplete. Evaluation practice, although improved, remains insufficiently standardized. Security, watermarking, consent, and anti-spoofing must keep pace with synthesis quality. For practitioners, zero-shot voice cloning is increasingly usable in controlled real-world conditions. For researchers, the most compelling questions now concern how to make synthetic speech not only realistic, but also controllable, safe, equitable, and reliable under the messy conditions of actual deployment.

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CONFLICT OF INTEREST

The authors declare no conflict of interest.

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